Chapter #6

Bitcoin Wallets Unlocked: Navigating Self-Custody and the Lightning Network for Secure Transactions

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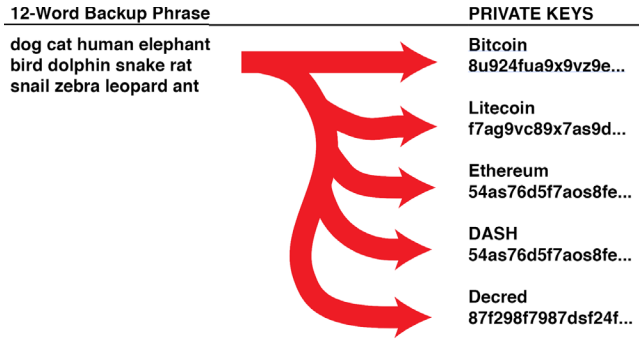
6.0 From Novice to Pro: Navigating the World of the Bitcoin Wallet

When Sats are purchased for the first time, they will be credited to a virtual account, similar to how funds are deposited into a bank account.

The key difference is that while a bank account is centralized and subject to government regulations, a **bitcoin wallet** is decentralized and operates on a **person-to-person** network.
• ***Bitcoin*** has no central point of failure, but it is important to be cautious as someone's ***bitcoin*** can be in the possession of a third party, who is managing it.

• This virtual account, often referred to as a "wallet," is protected by a ***master private key***, much like how a bank account is protected by a *personal PIN or password*. Just as you have control over the funds in your bank account, you can control the Sats in your wallet, and use them to make purchases or transfer them to other accounts.

• Just as a locksmith can create any number of keys that can be used to open locks, a ***recovery or backup phrase*** (*or master private key*) can be used to generate any number of **private keys** that can be used to access your Bitcoin wallet. You could say that a ***recovery phrase*** is like a locksmith, and the **private keys** are the keys that are created by the locksmith.

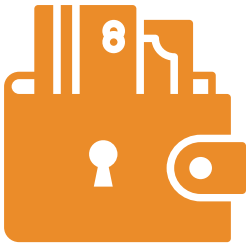


This table includes the two main types of bitcoin wallets, **self-custodial** and **custodial**. You can see the benefits and risks of using each wallet type and who controls the bitcoin in each case. Self-custodial means the user holds the **private keys**, *which means they are in true possession of their **bitcoin***, while with the second type, the *third party holds them*.

Wallet Type	Who controls my bitcoin?	Benefits	Risks
Self-Custodial Wallets	The user	Complete control over funds and transactions, no approval process or account freeze, no corporate or government control, protected against arbitrary confiscation, like keeping money at home.	No recovery if recovery phrase is lost, less customer support, full responsibility falls on the user.
Custodial Wallets	The third-party provider	Easy recovery if access is lost, easier customer support	Funds are always connected to the internet, more vulnerable to hacking and breaches.

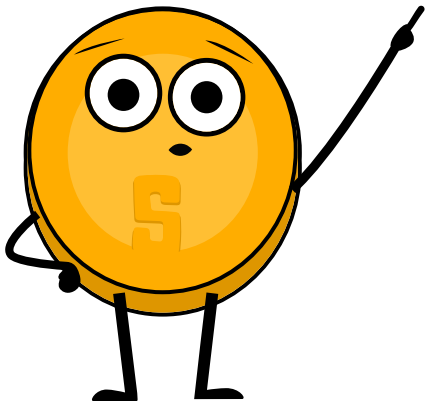
In a **self-custodial wallet** (also called **non-custodial wallet**,) you are the only one with the **keys** to the wallet and **you have full control over what goes in and out**. On the other hand, in a custodial wallet, someone else holds the key and can access and manage the contents of the wallet on your behalf.

- Self-custody is like being your own bank. Transactions are not subject to control or authority by any government or company, but it also means you bear full responsibility for keeping your ***bitcoin*** secure.
- Self-custody ensures that third parties cannot confiscate your ***bitcoin*** without your consent.
- Self-custody gives peace of mind in times of uncertainty, knowing your ***bitcoin*** is secure.



It's important to choose the right type of wallet for each individual's needs. Sometimes people find it hard to distinguish whether they are installing a custodial or a self-custodial wallet. This table shows the differences in installation process.

Wallet Type	Step 1: Choose a Wallet	Step 2: Install the Wallet	Step 3: Create a New Wallet	Step 4: Secure Your Seed Phrase	Step 5: Start Using Your Wallet
Self-Custodial Wallets	Choose a self-custodial wallet provider	Follow the wallet provider's instructions	Generate ***recovery phrase*** and at least one **private key**	Store the ***recovery phrase*** in a secure location	Start using the wallet to receive and send ***bitcoin***
Custodial Wallets	Choose a custodial wallet provider	Follow the wallet provider's instructions	Create an account with the wallet provider	N/A (wallet provider holds the **private keys**)	Start using the wallet to receive and send ***bitcoin***



When it comes to storing your ***bitcoin***, it's not just about who has control over it - there are many other risks to consider as well. That's why it's important to find a storage solution that is both secure and convenient.

Wallet Type	Description	Advantages	Disadvantages	Example User
Online Wallet	A wallet that is accessed through a web browser.	Accessible from any device with an internet connection. Easy to use.	Less secure. Can be hacked or compromised.	Someone who needs to access their wallet frequently and doesn't have a lot of funds to store.
Mobile Wallet	A wallet that is installed on a mobile device.	Convenient. Can be accessed from anywhere.	Can be lost if the device is misplaced, stolen, or hacked.	Someone who needs to make transactions on the go and doesn't have a lot of funds to store.
Desktop Wallet	A wallet that is installed on a desktop computer.	More secure than online wallets. Can be used offline.	Can be hacked if the computer is infected with malware.	Someone who wants to store a large amount of ***bitcoin*** and is comfortable with using a desktop computer.
Hardware Wallet	A physical device that stores ***bitcoin*** offline.	Very secure. Can be used offline.	Funds could be unrecoverable if the device is lost or stolen.	Someone who wants to store a large amount of ***bitcoin*** and is willing to pay for the added security of a hardware wallet.
Paper Wallet	A physical record of a bitcoin wallet's private and public keys.	Very secure. Can be used offline.	Can be lost or stolen if the physical record is lost or stolen.	Someone who wants to store a large amount of ***bitcoin*** and is willing to take the added precautions to ensure its security.

Analyze the trade-offs of the wallets and know there is no ideal wallet that satisfies all needs.

- When choosing a bitcoin wallet, there are several things you should consider:
  - **Security**: Make sure the wallet has strong security measures in place, such as two-factor authentication and secure password policies.
  - **Privacy**: Consider whether the wallet allows you to remain anonymous, or if it requires personal information to set up an account.
  - **Ease of Use**: Choose a wallet that is easy to use and navigate, especially if you are new to using ***bitcoin***.
  - **Compatibility**: Make sure the wallet is compatible with your device and operating system.
  - **Fees**: Compare the fees charged by different wallets to make sure you are getting the best deal.

- **Reputation**: Research the reputation of the wallet and its team to make sure it is trustworthy.
- **Control**: Some wallets give you more control over your private keys, which can be a security advantage. Consider whether you want a wallet that gives you full control, or one that is more user-friendly but may have less control.

You can always transfer your funds to a different wallet later.

6.1 The Process of Onboarding and Securing Your bitcoin

Onboarding into ***Bitcoin*** refers to the process of acquiring and using ***bitcoin***.

Before moving any further, it's important that we learn the steps for **onboarding** and familiarize ourselves with the process for **buying** and **securing** ***bitcoin*** safely.

1. **Choose a bitcoin exchange or brokerage**: There are many different platforms that allow you to buy and sell ***bitcoin***. Choose a platform that meets your needs and is reputable.
2. **Create an account**: Follow the platform's instructions to create a new account. This may involve providing personal information and verifying your identity.
3. **Connect a payment method**: Most platforms will allow you to connect a bank account, credit card, or debit card to fund your account. Follow the platform's instructions to add your payment method.
4. **Place an order**: Once your account is set up and funded, you can place an order to buy ***bitcoin***. The platform will provide you with a price quote and you can specify the amount of ***bitcoin*** you want to buy.
5. **Confirm the transaction**: Review the details of your transaction and **confirm the purchase**. The platform will process the transaction and the ***bitcoin*** will be transferred to your account on the platform.
6. **Withdraw the** ***bitcoin***: If you want to transfer the bitcoin to a self-custodial wallet, you will need to *withdraw the bitcoin from the platform and send it to your wallet*. The platform will provide instructions for how to do this.

"Not your keys, not your coins"

This is a popular saying among ***bitcoin*** holders. It refers to the idea that if you don't have direct control over the private keys associated with your bitcoin wallet, you don't have true ownership of the coins.

The **private key** is a secret code that allows you to access your ***bitcoin*** and spend it. When you

store your ***bitcoin*** with a third-party service, like an exchange or online wallet, you are relying on that service to keep your private key safe. If the service is hacked or goes out of business, you could lose access to your ***bitcoin***.

So, the saying "**Not your keys, not your coins**" is a reminder that it's important to take control of your own **private keys** and store them securely. By doing this, you can ensure that you have full control over your ***bitcoin*** and can access it whenever you want.

NOT YOUR KEYS
NOT YOUR COINS

6.1.1 Class Exercise: Mastering Self-Custody and Using Your Wallet With Confidence

If students do not have cell phones, the teacher will provide one to each student to borrow. There are two options for this activity:

Class Exercise. Option 1. Download a new wallet. Guide students step by step:

How to create and use a bitcoin wallet.

1. Search for the app in the App Store (iOS) or Google Play Store (Android).

2. Open the app and type in your 12- or 24-word recovery phrase.
Be sure to write it down. Keep it in a safe place. Remember that if you lose or forget this sequence of words, you will not be able to access your bitcoin if you lose access to your wallet.

3. You must then **confirm** that you have actually saved your recovery or **seed phrase**. To do this, you must **enter**, in the same order, the **words** of your seed phrase.

4. As an additional measure of security, some wallets allow you to **choose** a secure password.
- Your **private key** and first bitcoin address are automatically created for you by your wallet.

5. Use your "**receive**" address to receive ***bitcoin***.
Transfer *bitcoin* to your wallet.
- With a self-custodial wallet, you cannot always buy ***bitcoin*** directly with fiat, so you might need to purchase and transfer it from an exchange first.

Class Exercise. Option 2. Restore Wallet (Time Limited).

Download a bitcoin wallet and add some sats for each student. Give each student a sheet with a seed phrase to retrieve a wallet. Guide students step by step:

1. When you first start your wallet, you will see three methods of wallet creation, tap **[Import an existing wallet]**
• You will see an introduction screen, tap **[Restore with recovery phrase]**

2. Enter your 12/18/24-word recovery phrase one by one, in the correct order.

3. Touch [**Restore/Restore**] when finished.

4. You will see an *"Import Successful"* mode when your wallet has been successfully imported.

6.1.2 Class Exercise: How do I Receive bitcoin (in detail)

To receive ***bitcoin***, you will need to provide the sender with your bitcoin wallet **address**. This is a unique string of letters and numbers that represents your wallet and is used to identify it on the ***Bitcoin Network***. You can find your wallet **address** by logging into your bitcoin wallet and looking for an option to "Receive" or "Deposit" ***bitcoin***.

You can then share your bitcoin **address** with the sender in one of several ways:

• **Copy and paste the address**: You can copy the **address** by highlighting it and pressing "Copy" on your keyboard, then paste it into an email or message to the sender.
• **Share a link to your bitcoin wallet**: Some bitcoin wallets allow you to create a link to your wallet that you can share with the sender. They can then click on the link to access your wallet and send the bitcoin.
• **Share a QR code**: If the sender has a smartphone with a bitcoin wallet app, they can scan the QR code to get your bitcoin **address**.

Once the sender has your bitcoin address, they can send you the ***bitcoin*** by entering your address and the amount they want to send you and initiate the transaction. The ***bitcoin*** will then be sent to your wallet and will be visible once the transaction is confirmed on the ***Bitcoin Network***. It usually takes a few minutes.

6.1.3 Class Exercise: How Do I Send bitcoin and Pay for Goods and Services (in detail)

To send ***bitcoin***, you will need a few things: a bitcoin wallet, the recipient's bitcoin **address**, and the amount of ***bitcoin*** you want to send.

1. Open your bitcoin wallet.

• An SMS code will be sent to your phone number, and you will need to enter it in the dialog box. Alternatively, if you have enabled Google 2FA, you will need to enter the six-digit code from the Google Authenticator app.

2. Navigate to the "Send" or "Withdraw" feature and copy the recipient's address.

3. Enter the recipient's ***bitcoin*** address by pasting it in the "To" field.

4. Enter the amount of ***bitcoin*** you want to send in the "Amount" field.

5. Double-check the recipient's address and the amount to be sent.

6. Before clicking **Confirm and Send**, we recommend that you double-check the transaction details one more time to ensure that you are sending the correct amount of ***bitcoin*** to the correct wallet address.

7. Confirm the transaction and wait for the network to confirm the transaction.

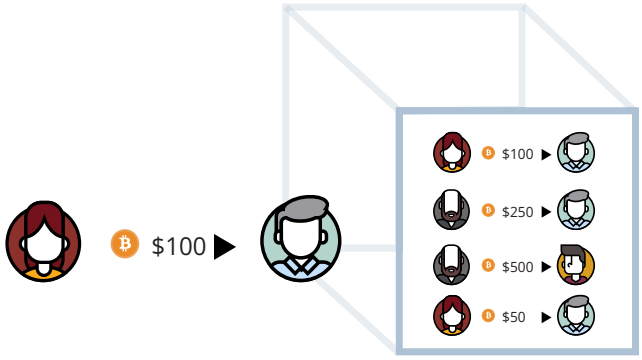
- Let's practice!!! Go to the coffee shop to **buy** snacks with ***bitcoin***.

6.2 On-Chain vs. Off-Chain

It is important to note that not all ***bitcoin*** transactions are recorded on the main ***Bitcoin blockchain***, some networks use different *blockchains* called sidechains to record transactions.

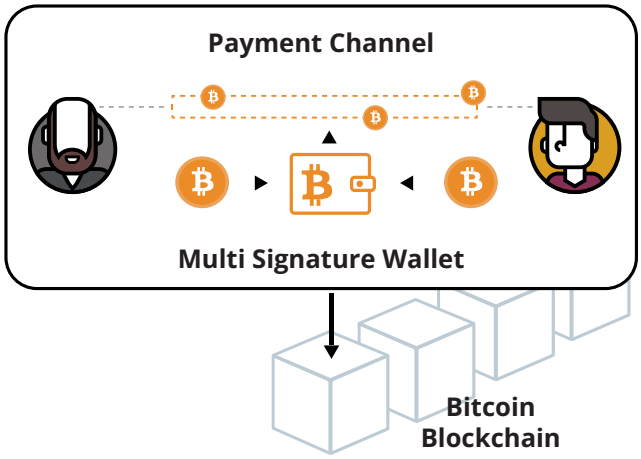
On-chain transactions:

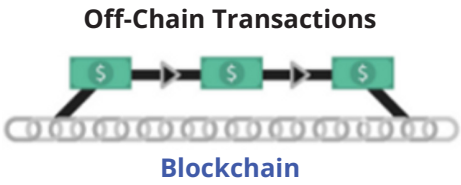
- These are transactions that happen directly on the ***Bitcoin blockchain***.
- They take about 10 minutes to confirm and the fees depend on the size of the transaction in bytes.
- They are secure but can be slower.



Off-chain transactions (Lightning Network):

- These transactions happen on a separate network built on top of the ***Bitcoin blockchain***.
- They are settled faster and with lower fees.
- They are commonly used where regulations and laws support their adoption and where the speed and cost of transactions are more important.
- When compared to transactions on-chain, they are less secure.





The Lightning Network is a scaling **approach** to ***Bitcoin***. It's really about moving a lot of ***Bitcoin*** transactions **out of the blockchain and into private channels** between users, but still counting on the blockchain's security.

If using the lightning network, only **three types of transactions** need to be broadcast to the *blockchain*.

The transaction that **opens** a channel.

The transaction that **closes** a channel.

Disputes that need to be settled.

Payment Network	Bitcoin Network	Lightning Network
Definition	A decentralized digital network that uses cryptography to secure financial transactions.	A second layer payment protocol that operates on top of the ***Bitcoin blockchain***, enabling faster and cheaper transactions.
Advantages	- Decentralized and secure - No chargebacks or fraud - Can be used anonymously - Global Acceptance	- Faster and cheaper transactions - Increased scalability - Off-chain transactions do not clog the blockchain
Disadvantages	- Slow transaction times - High fees for certain types of transactions- Complex for beginners	- Requires trust in the channel operators - Still experimental and not widely adopted - Requires on-chain transaction to open and close channels

6.3 The Lightning Network

Bitcoin is known for its unalterable public ledger, but it may not be the best choice for everyday transactions like buying coffee. The process of broadcasting these transactions to many nodes and storing them in a shared database can be slow and cumbersome. For personal or private transactions, it's better to use peer-to-peer payment channels.

A better solution is a layered approach to scaling, such as the combination of ***Bitcoin*** and Lightning Networks. This allows users to pick the layer that fits their needs. ***Bitcoin*** is a digital currency that is decentralized, while the Lightning Network provides quick, cheap, and confidential payments.

The Lightning Network is a payment system that allows users to send and receive payments quickly and inexpensively using ***bitcoin***. It works by setting up a shared wallet where both people store their ***bitcoin***, and then making unlimited transactions between each other without touching the main *blockchain*. When they're done, the final balance is recorded on the main *blockchain*.

Lightning operates as a separate network connected to the ***Bitcoin blockchain*** and is designed to work seamlessly with ***Bitcoin***. Taro, which is a recent addition to Lightning, now enables the network to be used for other types of assets, such as stablecoins, allowing users to make nearly instantaneous, low-cost payments in a currency linked to traditional fiat such as the US dollar. Payments can be made directly to the recipient, bypassing intermediaries, and converting the payment into the original currency before it reaches the store.

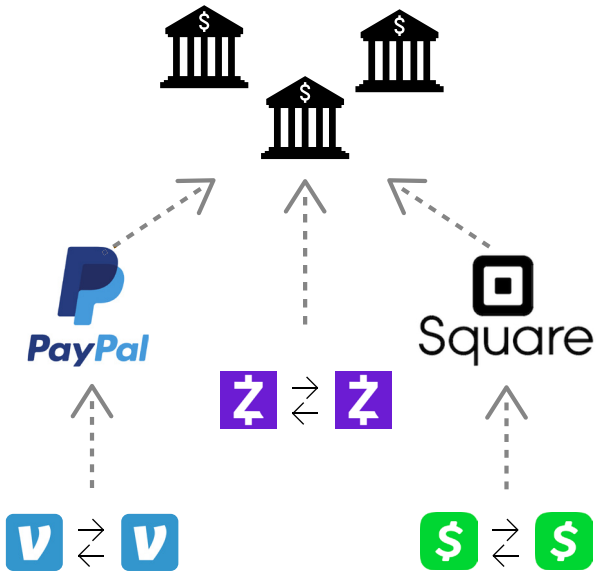
Using stablecoins in the Lightning Network for international transactions, such as remittances, provides several benefits:

1. Reduced costs: Cross-border transactions can be expensive due to fees charged by banks or other intermediaries. By using stablecoins in the Lightning Network, these fees can be reduced or eliminated, making cross-border payments more affordable.

2. Increased speed: Cross-border transactions can take several days to complete when using

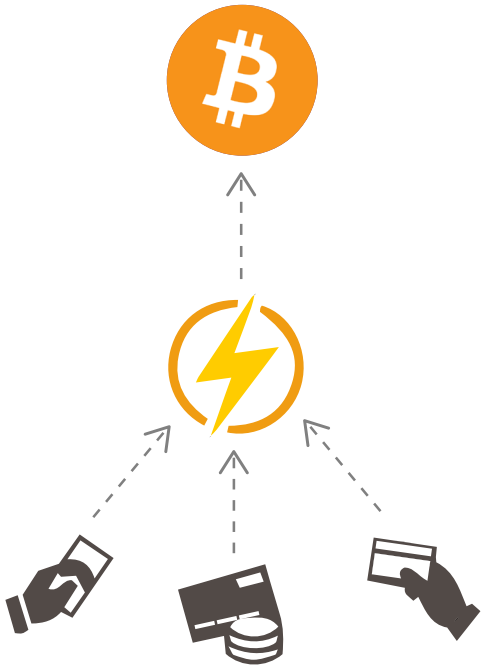
Modern Monetary System = Closed Networks

Banks Maintain Finality



Bitcoin Monetary System = Open Network

Bitcoin Maintains Finality



The Lightning Network provides the benefits of digital wallets like Apple Pay without the price volatility associated with ***bitcoin***.

traditional methods. By using stablecoins in the Lightning Network, international transactions can be processed quickly, reducing the time required to complete the transaction.

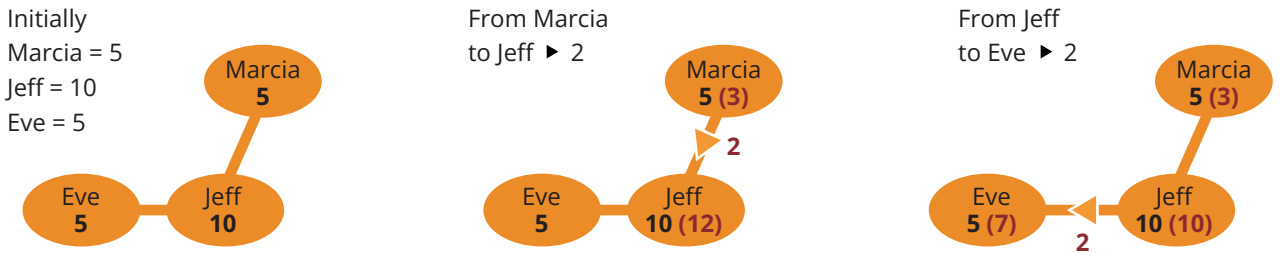
3. Improved access: For individuals or businesses in countries with limited access to traditional banking services, the use of stablecoins in the Lightning Network can provide a means to make international payments, thereby improving access to financial services.

6.3.1 A Lightning Transaction

Example#1

• Below, Marcia has 5 units of some currency and Eve has 5 units as well. Marcia wants to send 2 of her units to Eve, so she sends 2 units to Jeff. Jeff then passes on the 2 units to Eve, who now has 7 units. Marcia now has 3 units. And that's it! The transaction is done.

The key point here is that Marcia and Eve don't have to go through a bank or other intermediary to make the transaction happen.



• Jeff acts as an intermediary or a "**trusted third party**" in this scenario, where Marcia and Eve do not trust each other directly. Jeff receives the 2 units from Marcia and then passes it on to Eve, thus completing the transaction. By using Jeff as an intermediary, Marcia and Eve can complete the transaction without the need for a bank or other centralized institution, which can make the transaction faster, cheaper, and more secure. Jeff is a key element in this peer-to-peer transaction process

As a node operator in a Lightning Network transaction, Jeff benefits in several ways:

1. Transaction fees: Jeff earns a small fee for each transaction that passes through his node, which compensates him for the time and effort he puts into maintaining and running his node.

2. Network participation: By running a Lightning node, Jeff is participating in the network and helping to increase its decentralization, security, and stability. This can increase Jeff's reputation and credibility as a reliable node operator, making him a more attractive intermediary for future transactions.

3. Network growth: As the Lightning Network grows and more people use it, the number of transactions passing through Jeff's node is likely to increase, which can result in increased income from transaction fees.

4. Increased network security: Jeff's role as an intermediary helps to increase the security of the network by adding an additional layer of protection between Marcia and Eve. This can increase the confidence of users in the network, making it more attractive to new users and helping to drive growth.

Overall, being a node operator in the Lightning Network can provide Jeff with a steady source of income, as well as the opportunity to contribute to the growth and development of the network.

In summary, On-chain transactions are slower but more secure while off-chain (Lightning Network) are faster but less secure. You should consider the tradeoff between security and speed depending on your needs.

Example#2

Mina has a serious love for McDonald's. She goes there for breakfast, lunch, and dinner every day! But with so many different payment options available, she's not sure which one is the best choice. Luckily, she's learned a little bit about ***Bitcoin*** and The Lightning Network. After comparing the tables below, Mina has no doubt that using a Lightning Payment method is the way to go.

Benefits	Lightning	Traditional Banking System
Speed	Fast	Slow
Transparency	Transparent	Opaque
Security	Secure	Vulnerable
Transaction Fees	Low	High
Financial Inclusion	High	Limited

Benefits	Lightning	On-Chain
Scalability	High	Low
Privacy	High	Moderate
Interoperability	High	Low
Legal Compliance	Moderate	High
Cost-effectiveness	High	Moderate

Visa, Inc.

On average 1,700 transactions per second.

Capacity of 65,000 transactions per second.

Bitcoin On-chain

Capacity of 7 transactions per second.

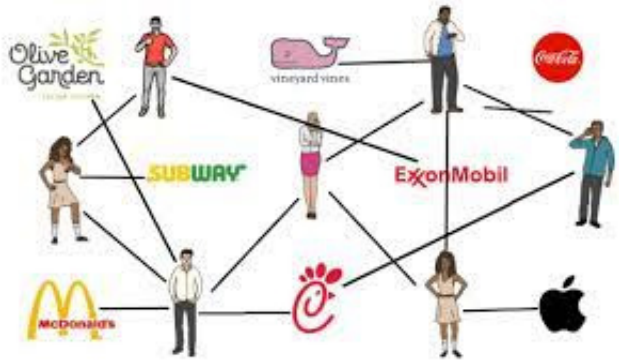
Bitcoin Lightning Network

Millions of transactions per second.

Mina is also a fan of fast, secure, and cost-effective transactions, so she decided to use Lightning for her purchases at McDonald's. With Lightning, she can enjoy her meals even more knowing that her payments are processed instantly, securely, and with low fees. Plus, since the Lightning Network provides financial inclusion, Mina can now pay for her meals even if she is in a remote area in El Salvador.

To get started with Lightning, Mina first downloads a Lightning Wallet on her phone. She then funds her Lightning Wallet by sending some bitcoin from her regular bitcoin wallet to her new Lightning Wallet. This process is called "funding the wallet" or "funding a payment channel." Mina can fund her wallet with any amount of bitcoin she is comfortable with, but it's important to note that the amount of bitcoin she locks in her Lightning Wallet cannot be used in her on-chain transactions.

Once her Lightning Wallet is funded, she can use it to make payments to McDonald's. McDonald's has a Lightning Node, so Mina can open a payment channel with them by sending some of her bitcoin from her Lightning Wallet to a specific address provided by McDonald's. This moves her ***bitcoin*** from the Bitcoin blockchain to an off-chain transaction on the Lightning Network.



With the payment channel open, Mina can now make purchases at McDonald's without having to open a new channel or pay high fees each time. The channel stays open as long as both Mina and McDonald's want to use it. For example, if Mina buys a hamburger for 0.0005 ***bitcoin***, the channel tracks that Mina now has 0.9995 ***bitcoin***. And if she buys a milkshake for 0.0003 ***bitcoin*** the next day, the channel tracks that Mina now has 0.9992 ***bitcoin***.

When Mina decides she wants to use her ***bitcoin*** balance for something else, she closes the channel by broadcasting a closing transaction to the ***Bitcoin*** *blockchain*. This is done by initiating a closing transaction in her Lightning Wallet, and the transaction contains the final balance of the channel agreed to by both parties. The transaction is then broadcast to the ***Bitcoin*** *blockchain* and confirmed by a miner. Once the transaction is confirmed, the channel is closed, and the remaining ***bitcoin*** in the channel will be returned back to Mina and McDonald's.

It's important to note that closing a channel can take some time to be confirmed on the *blockchain*. During this waiting period, the funds are still locked in the channel and cannot be used for on-chain transactions. Mina will receive a notification once the closing transaction is confirmed.

6.3.2 Class Exercise: Lightning Wallet Relay Race

1. First, you will need to download a Lightning wallet onto your phone or computer. There are several options to choose from, including Muun, Blue Wallet, Bitcoin Beach Wallet, and Eclair are a few options for mobile phones, and Lightning App and Zap for desktop computers.

2. Follow the instructions for installing the wallet on your device. This may involve downloading the app from the App Store or Google Play, or downloading and installing the software from the wallet's website.

3. Once the wallet is installed, open it and follow the prompts to set it up. This may involve creating a new wallet or restoring an existing one, and securing it with a password or other form of authentication.

4. Make sure that you have a way to receive satoshis. This may involve providing your wallet with a receiving address, or scanning a QR code provided by your teacher or another member of your group.

5. When your wallet is set up and you are ready to receive satoshis, your teacher will give you and your group a starting amount of satoshis by sending them directly to your wallet.

A. Your group's goal is to pass the satoshis from one person's wallet to another, using the Lightning Network, until they reach the last person in the group.

B. To send satoshis to another person, open your wallet and follow the instructions for making a payment. You will need to provide the recipient's wallet address or scan a QR code, and enter the amount of satoshis that you want to send.

C. If your group is the first to successfully send the satoshis to the last person, you win! (And get to keep the sats, and some candy.)

6.3.3 Class Exercise: Lightning Online Interactive Demo

Class Exercise. Begin by exploring one of the interactive websites provided by the teacher. Then, follow the instructions in the next page.

- ***https://lnrouter.app/graph/zero-base-fee***

- ***https://www.robtex.com/lnemulator.html?conf=A5-5B,B5-5C&send=A2C***

1. Focus on the key concepts discussed in class, including payment channels, routes, and fees.

2. Take note of any questions or difficulties you encounter as you explore the website.

3. Work with your group to share your findings and discuss any questions or difficulties with the class.

4. Be prepared to participate in class discussions about the Lightning Network and its potential as a scaling solution for bitcoin transactions.